Hugo Gaquere

High-Performance & Real-Time Software Engineer (GPU · CPU)

PhD candidate, Observatoire de Paris – Université PSL · defense Sept. 2027

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Software engineer specialized in high-performance and real-time computing on GPUs and CPUs, finishing a PhD on real-time processing of large data volumes for radio telescopes, and for radar systems in collaboration with Thales LAS. I build C++/CUDA software end to end, from performance analysis to integration in production pipelines.

Experience

PhD Researcher: real-time GPU/CPU processing for radio telescopes and radar Sept. 2024 – Sept. 2027
Observatoire de Paris – PSL (LIRA) Meudon, France

- Designed and built **fast-deconv**, an open-source C++ library with **CUDA** and **OpenMP** backends that accelerates state-of-the-art **deconvolution**, the most expensive step of radio-telescope image reconstruction.
- Optimized for memory throughput: performance analysis shows the main GPU kernels reach **74–85% of peak memory bandwidth**; this efficient GPU mapping of deconvolution opens the way to real-time imaging.
- Validated on telescope data (**400-megapixel images**), matching the output of the reference software; working with astronomers to integrate the library into their production pipeline.
- With **Thales LAS**: developing a high-performance GPU pipeline for **real-time** sparse deconvolution and high-resolution imaging of radar signals.

Teaching Assistant: GPU Programming with CUDA 2027 (upcoming)
ESIEA Paris, France

AI Developer Intern: maritime target detection & tracking Apr. – Sept. 2023
MaxSea International Bidart, France

- Trained object-detection models with an MLOps pipeline and built a .NET library for inference and target tracking.
- Optimized inference on CPU and integrated GPU with ONNX Runtime and OpenVINO.

Earlier experience

Master's project: hornet-nest detection software for BeesForLife (Python, Qt, computer vision) 2023
Web developer, 1% immobilier: real-estate agency website (React, Next.js, Firebase) 2022
Mobile developer intern, MINK (Kotlin, Android) 2020

Publications

H. Gaquere, C. Tasse, O. Aumage, O. M. Smirnov, F. Martens, D. Gratadour. *A high-performance GPU accelerated framework for radio-interferometric deconvolution.* Proc. SPIE 14153, 2026 (paper and poster). doi:10.5281/zenodo.23009533

Skills

HPC & GPU	CUDA, OpenMP, HPC clusters (Slurm), performance analysis & optimization
Domains	Real-time signal processing, image processing, deconvolution & inverse problems
Programming	C++, CUDA, Python, C, C#, Bash
Tooling	Git, CMake, Conan, CI/CD (GitHub Actions), GoogleTest, Linux
ML & vision	PyTorch, ONNX Runtime, OpenVINO, OpenCV, object detection
Languages	French (native), English (professional working proficiency)

Education

PhD in Astronomy & Astrophysics, Observatoire de Paris, Université PSL (adv. D. Gratadour, L. Matias) 2024 – 2027
Thesis: Enabling technologies for real-time acquisition and processing of large data volumes, applied to giant astronomical telescopes and radar systems.

MSc in Computer Science (image & sound processing), Université de Bordeaux 2023
BSc (2021) and **DUT** (2020) in Computer Science, Université de Bordeaux 2018 – 2021